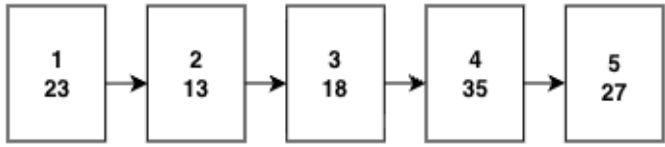
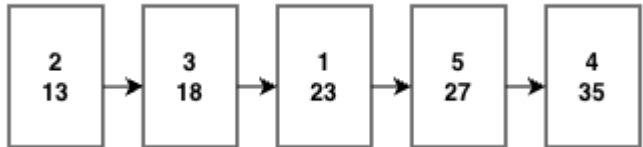


Tab 1

Question#1 [Coding] [12 Marks]

Before boarding, the airline staff checks the passengers one by one to make sure they are arranged properly according to their bag weights. Starting from the first passenger, each passenger is compared with the passenger standing immediately next to them in the list. **If the current passenger's bag weight is greater than the next passenger's bag weight, their data values are swapped by exchanging the passenger_id and bag_weight.** Otherwise, they remain in the same position. This process continues by moving one passenger forward each time until reaching the **second-to-last** passenger, since the last passenger has no next passenger to compare with.

Assume, the passengers are stored in a **singly linked list**, where each node contains: **passenger_id**, **bag_weight** and **next**. Write a function that takes the **head** of the linked list and performs this rearrangement. **Note: Creating a new Linked List is NOT ALLOWED.**

Java Custom Node Class (Given)	Python Custom Node Class (Given)
<pre>class Node { int passenger_id; int bag_weight; Node next; Node(int passenger_id, int bag_weight) { this.passenger_id = passenger_id; this.bag_weight = bag_weight; this.next = null; } }</pre>	<pre>class Node: def __init__(self, passenger_id, bag_weight,next): self.passenger_id = passenger_id self.bag_weight = bag_weight self.next = next</pre>
Java Method Signature	Python Method Signature
<pre>public static void rearrangePassengers(Node head){ //To do }</pre>	<pre>def rearrange_passengers(head): # To do</pre>
Sample Given Linked List	Sample Output
	
Explanation: → (1,23) vs (2,13) → swap → (2,13)(1,23)(3,18)(4,35)(5,27) → (1,23) vs (3,18) → swap → (2,13)(3,18)(1,23)(4,35)(5,27) → (1,23) vs (4,35) → no swap → (4,35) vs (5,27) → swap → (2,13)(3,18)(1,23)(5,27)(4,35)	

Question#2 [Short Questions] [6 marks]

i) In a 2D array declared as `int arr[3][4];` How many elements are there?
[1 mark]

- (A) 7 (B) 12 (C) 24 (D) 3

ii) Which of the following operations is NOT possible directly in a singly linked list? [1 mark] A) Traversing from head to tail B) Deleting the first node C) Inserting a node after a given node D) Accessing the previous node of a given node		iii) What will happen if the head pointer of a singly linked list is NULL? [1 mark] A) The list is empty B) The list contains one node C) The list is circular D) The list has garbage values
iv) What is the time complexity for the following code? [3 marks]		
Java	Python	ANSWER BELOW
<pre>int sum = 0; for (int i = 1; i < n; i *= 2) { for (int j = 0; j < i; j++) { sum++; } }</pre>	<pre>s = 0 i = 1 while i < n: for j in range(i): s += 1 i *= 2</pre>	

You can start coding [Question#1](#) from here: